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Method and apparatus to create a virtualized replica of a computer network

Abstract

Disclosed herein are system, method, and computer program product aspects for generating a virtual replica of a physical network. An aspect operates by receiving, from a network management system (NMS), a request to generate a network digital twin (network-DT) instance corresponding to a physical network that includes a plurality of physical network devices, where the request includes link connectivity information corresponding to the physical network and a respective plurality of device characteristics corresponding to each of the plurality of physical network devices. A plurality of device digital twin (device-DT) instances are generated, each corresponding to a respective physical network device of the plurality of physical network devices, where each device-DT instance of the plurality of device-DT instances is generated based on the respective plurality of device characteristics corresponding to the respective physical network device. The network-DT instance is generated based on the plurality of device-DT instances and the link connectivity information.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5946472	12/1998	Graves et al.	N/A	N/A
10320625	12/2018	Cherumbath	N/A	N/A
11012566	12/2020	Velusamy	N/A	N/A
11863450	12/2023	Obediente et al.	N/A	N/A
2006/0253588	12/2005	Gao et al.	N/A	N/A
2015/0067143	12/2014	Babakhan et al.	N/A	N/A
2016/0364201	12/2015	Beveridge et al.	N/A	N/A
2020/0328776	12/2019	Scholand et al.	N/A	N/A
2020/0382412	12/2019	Chandrappa et al.	N/A	N/A
2020/0389373	12/2019	Barton	N/A	H04L 41/0895
2021/0306416	12/2020	Mukhopadhyay	N/A	H04L 41/0846
2022/0100647	12/2021	Hamid	N/A	N/A
2022/0200917	12/2021	Mortensen	N/A	H04L 41/145
2022/0237243	12/2021	DeLuca et al.	N/A	N/A
2024/0037020	12/2023	Hamid	N/A	N/A
2024/0095158	12/2023	Miriyala et al.	N/A	N/A
2024/0112506	12/2023	Smith	N/A	H04L 41/0895
2024/0160466	12/2023	Yuan	N/A	H04L 41/0895
2024/0195683	12/2023	Wang	N/A	H04L 41/0823
2024/0406277	12/2023	Henkel et al.	N/A	N/A
2025/0004463	12/2024	Ezrielev	N/A	G05B 23/0275

OTHER PUBLICATIONS

VMware's vCenter Converter, https://www.vmware.com/products/conver, downloaded Jul. 14, 2023; 6 pages. cited by applicant

U.S. Appl. No. 18/362,259, filed Jul. 31, 2023; 64 pages. cited by applicant

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Background/Summary

BACKGROUND

(1) Computer network management involves configuring, monitoring, updating, and troubleshooting network devices. Network management systems facilitate the centralized management of network devices, such as routers, switches, firewalls, and servers. They allow network administrators to configure and control network devices, manage updates, and enforce standardized configurations across the network.

(2) Validating changes to a network, such as device configuration updates or the rollout of new applications and devices, prior to deployment is essential to reduce the risk of introducing disruptions or issues in the production network. However, modern networks can be highly complex, consisting of numerous devices, protocols, and interconnected systems. Validating configuration changes in such intricate environments can be challenging due to the sheer scale and interdependencies involved.

SUMMARY

(3) Some aspects of this disclosure relate to apparatuses and methods for creating a virtual replica of a computer network. For example, some aspects of this disclosure relate to using network device parameters and link connectivity information to customize a virtual-replica template and instantiate a digital twin instance of a physical network.

(4) Some aspects of this disclosure relate to a network system with a memory and a processor coupled to the memory. The processor receives, from a network management system (NMS), a request to generate a network digital twin (network-DT) instance corresponding to a physical network including a plurality of physical network devices. According to some aspects, the request includes link connectivity information corresponding to the physical network and a respective plurality of device characteristics corresponding to each of the plurality of physical network devices. The processor then generates a plurality of device digital twin (device-DT) instances, each corresponding to a respective physical network device of the plurality of physical network devices, where each device-DT instance of the plurality of device-DT instances is generated based on the respective plurality of device characteristics corresponding to the respective physical network device. The processor then generates the network-DT instance based on the plurality of device-DT instances and the link connectivity information, and a management channel is established between the network-DT instance and the NMS.

(5) According to some aspects, the processor receives, over the management channel, a request to modify configuration of the network-DT from the NMS, where the request to modify configuration includes a set of modified configuration parameters. The processor then updates the configuration of the network-DT instance based on the modified configuration parameters, and synchronizes the updated configuration of the network-DT instance and a running configuration of the physical network device using the management channel. According to some aspects, the configuration of the network-DT includes link connectivity information corresponding to the physical network. According to some aspects, the processor validates control plane, management plane, and data plane operations using the network-DT with the updated configuration.

(6) According to some aspects, to generate a device-DT instance of the plurality of device-DT instances the processor selects a virtual-replica template from a plurality of virtual-replica templates based on a first plurality of device characteristics of the respective plurality of device

characteristics. The processor then customizes the virtual-replica template based on the second plurality of device characteristics of the respective plurality of device characteristics, and generates the device-DT instance of the physical network device based on the customized virtualized-replica template.

(7) Some aspects of this disclosure relate to a non-transitory computer-readable medium (CRM) having instructions stored thereon that, when executed by at least one computing device, causes at least one computing device to perform various operations. According to some aspects, the operations include receiving a request to generate a network digital twin (network-DT) instance corresponding to a physical network including a plurality of physical network devices, where the request includes link connectivity information corresponding to the physical network and a respective plurality of device characteristics corresponding to each of the plurality of physical network devices. The operations further include generating a plurality of device digital twin (device-DT) instances, each corresponding to a respective physical network device of the plurality of physical network devices, where each device-DT instance of the plurality of device-DT instances is generated based on the respective plurality of device characteristics corresponding to the respective physical network device. The operations further include generating the network-DT instance based on the plurality of device-DT instances and the link connectivity information, and establishing a management channel between the network-DT instance and the NMS.

(8) This Summary is provided merely for purposes of illustrating some aspects to provide an understanding of the subject matter described herein. Accordingly, the above-described features are merely examples and should not be construed to narrow the scope or spirit of the subject matter in this disclosure. Other features, aspects, and advantages of this disclosure will become apparent from the following Detailed Description, Figures, and Claims.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) The accompanying drawings, which are incorporated herein and form part of the specification, illustrate the present disclosure and, together with the description, further serve to explain the principles of the disclosure and enable a person of skill in the relevant art(s) to make and use the disclosure.

(2) FIG. **1** illustrates an example network system for generating a virtualized replica of a computer network, according to some aspects of the disclosure.

(3) FIG. **2** illustrates an example exchange of communications that may occur in a network system to create a network digital twin, according to some aspects of the disclosure.

(4) FIG. **3** illustrates an example exchange of communications that may occur for synchronizing configuration changes between a network digital twin and a corresponding physical network, according to some aspects of this disclosure.

(5) FIG. **4** illustrates an exemplary method to generate a device digital twin instance of a physical network device, according to some aspects of this disclosure.

(6) FIG. **5** illustrates an exemplary method to generate a network digital twin instance of a physical network, according to some aspects of this disclosure.

(7) FIG. **6** is an example computer system for implementing some aspects or portion(s) thereof.

(8) The present disclosure is described with reference to the accompanying drawings. In the drawings, generally, like reference numbers indicate identical or functionally similar elements. Additionally, generally, the left-most digit(s) of a reference number identifies the drawing in which the reference number first appears.

DETAILED DESCRIPTION

(9) Provided herein are system, apparatus, device, method and/or computer program product

aspects, and/or combinations and sub-combinations thereof, for creating a virtual replica of a computer network. For example, aspects herein describe generating a digital twin (DT) instance of a network (e.g., a computer network) on a cloud network.

(10) Validating network and configuration changes prior to deployment reduces the risk of introducing disruptions at runtime. A physical sandbox environment can be used for validating the effectiveness of upgrades, such as new hardware, software updates, or configuration changes, and verifying their impact on network performance. A sandbox network is a separate and isolated network environment that closely resembles the production network but is used specifically for testing and troubleshooting.

(11) The sandbox network should mirror the production environment as closely as possible to ensure accurate network testing and validation. Creating a replica of the production network in the sandbox environment involves setting up similar network devices, servers, applications, and connectivity configurations. In addition, existing network device configurations also should be replicated from the production environment. However, duplicating network devices to create a physical sandbox network that resembles the production network can be an arduous, time-consuming process, and potentially cost prohibitive.

(12) To solve the above technological problem, aspects of this disclosure describe creating a DT instance of a physical network (herein referred to as "network-DT") that provides cost-effective network configuration assurance and network modeling. Furthermore, the network-DT includes multiple device-DTs, where each device-DT is a DT instance corresponding to a network device of the physical network. According to some aspects, the network-DT and/or the device-DT can be created automatically with no (or minimum) involvement from a user (e.g., a network administrator).

(13) FIG. **1** illustrates an example network system **100** for generating a virtualized replica of a computer network, according to some aspects of this disclosure. Network system **100** includes network management system (NMS) **102**, physical network **104**, virtual cloud network **108**, and client device **114**. Physical network **104** includes one or more physical network devices **106***a*-**106***e*. Furthermore, virtual cloud network **108** includes network digital twin (network-DT) **116** which is a virtual replica of physical network **104**. Network-DT **116** includes one or more device digital twins (device-DTs) **110***a*-**110***e*. According to some aspects, each of the device-DTs **110***a*-**110***e* is a virtual replica of one of the physical network devices **106***a*-**106***e*. Hence, each device-DT has a counterpart physical network device in the physical network **104**. Alternately, a device-DT may also be a virtual replica of more than one physical network device **106***a*-**106***e* (e.g., two or more network switches in a stacked configuration).

(14) According to some aspects, each of the physical network devices **106***a*-**106***e* may be a router, switch, bridge, network gateway, server, host, and/or the like. According to some aspects, one or more physical network devices (e.g., one or more of the physical network devices **106***a*-**106***e*) may be a wired client and/or a wireless client. Furthermore, one or more physical network devices (e.g., one or more of the physical network devices **106***a*-**106***e*) may be a wireless local area network (WLAN) access point and/or a WLAN controller. According to some aspects, hardware attributes of a physical network device **106***a*-**106***e* may include model name, serial number, MAC addresses, device expansion module type, interface module type, stacking composition, and the like. According to some aspects, software attributes of a physical network device **106***a*-**106***e* may include network operating system (NOS) type, NOS version, feature license(s), and the like. Furthermore, each physical network device **106***a*-**106***e* may be configured with a plurality of configuration parameters that include virtual local area network (VLAN) parameters, spanning tree protocol (STP) parameters, open shortest path first (OSPF) parameters, fabric parameters, virtual router redundancy protocol (VRRP) parameters, and the like.

(15) According to some aspects, network-DT **116** may be implemented by, for example, a data center including one or more servers in the cloud. According to some aspects, cloud network **108**

may include one or more physical devices and/or one or more applications hosted on a distributed computing platform, a cloud computing platform, a centralized hardware system, a server, a computing device, and/or an external network-to-network interface device, among others.

(16) According to some aspects, device-DTs **110***a*-**110***e* of network-DT **116** are virtual replicas of physical network devices **106***a*-**106***e*. Hence, the hardware attributes, the software attributes, and the device configuration of a device-DT and its corresponding physical network device may be identical. According to some aspects, a physical network device (e.g., one of the physical network devices **106***a*-**106***e*) and its corresponding device-DT in network-DT **116**, may be configured with the same network operating system (NOS). For example, physical network device **106***a* and device-DT **110***a* may be configured with the same NOS. For example, physical network device **106***c* and device-DT **110***c* may be configured with the same NOS. In some examples, physical network device **106***a* and physical network device **106***c* (and their corresponding device-DT) use different NOS. According to some aspects, when the physical network device is a host device, the corresponding device-DT instance in network-DT **116** may simulate and generate network traffic. Accordingly, a device-DT in network-DT **116** may have internet connectivity if the corresponding physical network device in physical network **104** has internet connectivity. According to some aspects, device-DTs **110***a*-**110***e* are configured with virtualized NOS instantiated in cloud network **108**.

(17) According to some aspects, network-DT **116** is a virtualized replica of the physical network **104**. Accordingly, the configuration of interconnections between device-DTs **110***a*-**110***e* mirrors the configuration of interconnections between physical network devices **106***a*-**106***e*. Since network-DT **116** mirrors the characteristics of physical network **104**, network-DT **116** provides a sandbox environment to validate the management plane, control plane, and data plane features of physical network **104**. According to some aspects, the use cases for which network-DT **116** can be utilized include, but are not limited to, validating new configuration and/or new features, validating software upgrades for devices in the network, scaled feature/device validations, agile customer demonstrations including various solution components, device/portfolio interoperability, network planning and best-practice applications, and the like. According to some aspects, the network-DT **116** instance replicates a physical device and is capable of forwarding network traffic based on the configured features.

(18) According to some aspects, NMS **102** manages the operation and function of physical network devices **106***a*-**106***e*. According to some aspects, NMS **102** may be a cloud-based network management system. Client device **114** may communicate with NMS **102** using an application programming interface (API) and/or a graphical user interface (GUI) to configure and manage physical network devices **106***a*-**106***e* and/or device-DTs **110***a*-**110***e*. Alternatively or additionally, NMS **102** may communicate with the physical network devices **106***a*-**106***e* using a wired link (e.g., coaxial cable, Ethernet cable, fiber optic cable, etc.), a wireless link and/or combinations thereof. According to some aspects, NMS **102** can be deployed as an on-premise solution (i.e., without deploying on an external cloud network). According to some aspects, NMS **102** may automate the management of physical network devices **106***a*-**106***e* and/or device-DTs **110***a*-**110***e* using a software defined networking (SDN) controller. According to some aspects, client device **114** may access NMS **102** via a password protected user interface. According to some aspects, NMS **102** may provide access to devices remotely using secure shell (SSH), secure socket layer (SSL)/transport layer security (TLS) protocols. According to some aspects, NMS **102** may support simple network management protocol (SNMP) protocol.

(19) Furthermore, NMS **102** may support full-stack management of physical network devices **106***a*-**106***e* and enable management operations such as onboarding, configuration, monitoring, troubleshooting, reporting, and the like. According to some aspects, a user can set up a network by onboarding physical network devices **106***a*-**106***e* onto the NMS **102**. Using NMS **102**, the user can configure and manage the network devices **106***a*-**106***e*. As an example, users can define global

settings, view device status, and customize device configurations at the device level. According to some aspects, using NMS **102**, users can assign devices to an existing user-defined location tree within the context of NMS **102**.

(20) According to some aspects, digital twin instantiation service (DTIS) **112** performs a variety of functions such as instantiating, updating, life-cycle management, deleting DTs, interconnection management of DT-devices, and auto horizontal scaling based on the size of physical network **104**. According to some aspects, DTIS **112** communicates with NMS **102** to exchange device characteristics and interconnectivity configuration corresponding to physical network devices **106*a*-106*e***. According to some aspects, in case a DT-device goes into erroneous state, DTIS **112** is responsible of recovering the DT-device either by shutting it down and relaunch it, recreating the links between the device-DT instances, or recreating a whole new instance with the same configuration. According to some aspects, when a DT-device goes into erroneous state, DTIS **112** recreates a new DT-device instance using a saved bundled configuration and/or a preserved disk from the previous instance. DTIS **112** can be implemented on virtual cloud network **108**. However, DTIS **112** can be co-located with NMS **102**, according to some aspects.

(21) According to some aspects, DTIS **112** maintains instances of virtualized NOS and virtual-replica templates. According to some aspects, DTIS **112** may store a plurality of virtual-replica templates corresponding to various types of physical network devices, such as routers, switches, bridges, network gateways, servers, hosts, and the like. According to some aspects, a virtual-replica template may define a superset of hardware and software attributes corresponding to a respective type of network device. For each type of physical network device, DTIS **112** may maintain a configurable virtual-replica template that can be configured based on a variety of hardware and software attributes. According to some aspects, a virtual-replica template can be configured and instantiated to generate a virtual replica of a corresponding physical network device

(22) FIG. **2** illustrates an example exchange of communications that may occur in a network system to create a digital twin of a network device, according to some aspects of this disclosure. In the illustrated example, the network system **200** includes a client device **114**, physical network devices **106*a*-106*e***, NMS **102**, DTIS **112**, and network-DT **116** that includes device-DTs **110*a*-110*e***.

(23) At **212**, client device **114** initiates creating a network twin instance using a user interface of NMS **102**. According to some aspects, client device **114** sends a request to NMS **102** to create a DT instance of physical network **104** on a cloud network (e.g., virtual cloud network **108**). In the example of FIG. **2**, network-DT **116** is the DT instance of physical network **104** which includes physical network devices **106*a*-106*e***. According to some aspects, client device **114** may communicate with NMS using an application programming interface (API). The request to generate a DT instance corresponding to physical network **104** may be sent as a REST API call.

(24) According to some aspects, NMS **102** may display a list of managed network devices in a physical network to a user of client device **114**. According to some aspects, the physical network **104** may be a large network with thousands of network devices spread over multiple buildings, multiple floors of a building, and/or over several campuses of an institution. According to some aspects, a graphical interface of NMS **102** may display a topological map of the physical network at user device **112**. According to some aspects, using NMS **112**, a user of client device **114** may select a portion of physical network **104** (e.g., physical network devices located on a single floor of a building or in a single campus) to replicate in cloud network **108**. According to some aspects, user of client device **114** may select the entire physical network **104** to replicate in cloud network **108**. A user of client device **114** may select a portion of the physical network **104** by drawing a boundary around a selected portion on the displayed topological map of the physical network. According to some aspects, a user may select a set of physical network devices in physical network **104** based on their serial numbers and/or model numbers. Additionally, the user may make a selection of a set of physical network devices in the physical network **104** based on their OS

persona, device model, and/or OS version.

(25) At **214**, NMS **102** sends a request to the physical network devices within the selected portion of the physical network to send details corresponding to their running configuration. According to some aspects, NMS **102** may communicate with physical a network device (e.g., physical network devices **106***a***-106***e*) using SSH, SSL/TLS, and/or SNMP protocols. According to some aspects, NMS **102** may send a <get> or a <get-config> remote procedure call (RPC) message to obtain a configuration bundle from a physical network device of physical network **104**. However, the aspects of this disclosure are not limited to these examples and can include other requests.

(26) At **216**, a physical network device (e.g., physical network device **106***e*) generates a bundle of its current configuration and/or its operating system characteristics, and sends it to NMS **102**. According to some aspects, configuration details sent by each of the physical network devices includes physical attributes of the device, device model, NOS version, and/or the like. According to some aspects, NMS **102** may actively manages all devices of physical network **104**, and it may have information corresponding to the hardware and software attributes of physical network devices **106***a***-106***e*. According to some aspects, client device **114** may obtain configuration details corresponding to one or more physical network devices (e.g., physical network devices **106***a***-106***e*) and their corresponding device-DTs from NMS **102** and store the configuration details (e.g., on local memory). At **216**, a physical network device (e.g., physical network device **106***e*) may load a previously saved configuration (e.g., configuration corresponding to physical network device **106***e*) and send it to NMS **102**.

(27) According to some aspects, NMS **102** may obtain link connectivity information from physical network devices **106***a***-106***e*. Physical network devices **106***a***-106***e* may learn layer **2** and layer **3** connectivity information through network discovery. According to some aspects, physical network devices **106***a***-106***e* may obtain link connectivity information using traceroute, simple network management protocol (SNMP) and/or link layer discovery protocol (LLDP). According to some aspects, using LLDP each physical network device (e.g., physical network devices **106***a***-106***e*) may obtain information corresponding to the neighboring devices. Neighbor device information that can be obtained using LLDP include device name and capabilities, MAC addresses, Port IDs, and management addresses. Using the LLDP information obtained from physical network devices **106***a***-106***e*, NMS **206** may create a topology map that provides a comprehensive view of network connectivity of the physical network **104**.

(28) At **218**, NMS **102** sends a request to DTIS **112** to create a DT instance of physical network **104** having physical network devices **106***a***-106***e*. According to some aspects, NMS **102** sends a request to DTIS **112** using a REST API call or a gRPC (gRPC remote procedure call). According to some aspects, the request to DTIS **112** includes a plurality of device characteristics corresponding to the physical network devices **106***a***-106***e*. According to some aspects, NMS **102** may send to DTIS **112** the link connectivity information corresponding to physical network devices **106***a***-106***e*. The link connectivity information sent by NMS **102** may include the identities of the directly connected neighboring devices for each physical network device and the ports that are used for connecting to the neighboring devices.

(29) According to some aspects, the request may include hardware attributes of physical network devices **106***a***-106***e*. Alternatively, the request may include both hardware attributes and software attributes of physical devices **106***a***-106***e*. According to some aspects, the request may include hardware attributes of physical network devices **106***a***-106***e*, such as model name, serial number, MAC addresses, device expansion module type, interface module type, stacking composition, and the like. According to some aspects, the request may include software attributes of physical network devices **106***a***-106***e*, such as NOS type, NOS version, feature license(s), and the like. Furthermore, the request includes the running configuration of physical each physical network device and/or configuration parameters such as VLAN parameters, STP parameters, OSPF parameters, fabric parameters, VRRP parameters, and the like. According to some aspects, at **218**,

NMS **102** sends information corresponding to device characteristics, including link connectivity information.

(30) At **220**, DTIS **112** initiates the instantiation of network-DT **116** having device-DTs **110***a*-**110***e*. According to some aspects, DTIS **112** may generate an instantiation of network-DT **116** using the device characteristics and the link connectivity information obtained from NMS **102**. DTIS **112** may store a plurality of virtual-replica templates and/or NOS images corresponding to various types of physical network devices. According to some aspects, each virtual-replica template may correspond to a specific hardware model of a specific type of physical network device (e.g. a specific model of a network switch). In addition, each virtual-replica template may define a superset of hardware and software attributes corresponding to a respective type of network device. In some aspects, DTIS **112** and network-DT **116** can be collocated (e.g., initially) on the same cloud network.

(31) According to some aspects, at **220**, DTIS **112** creates a device-DT corresponding to each of the physical network devices **106***a*-**106***e* in the physical network **104**. For example, DTIS **112** create instances of device DTs **110***a*, **110***b*, . . . , and **110***e* which are the digital replicas of physical network devices **106***a*, **106***b*, . . . , and **106***e*, respectively. According to some aspects, to create instances of device-DTs **110***a*-**110***e*, DTIS **112** may select a virtual-replica templates based on one or more hardware characteristics of the corresponding physical network devices. For example, to create device-DT **110***e*, DTIS **112** may select a predefined virtual-replica template based on one or more hardware characteristics of the physical network device **106***e*. According to some aspects, each virtual-replica template may have a memory requirement value based on the hardware model of the corresponding physical network device. According to some aspects, a virtual-replica template may specify predefined ports for communication between DTIS **112** and the corresponding device-DT of network-DT **116**, and/or for communication within network-DT **116**. DTIS **112** and/or NMS **102** may further customize the selected virtual-replica templates based on one or more software characteristics and/or hardware characteristics of physical network devices **106***a*-**106***e*. DTIS **112** may then instantiate the customized virtual-replica templates to generate device-DTs **110***a*-**110***e*. According to some aspects, the hardware characteristics based on which DTIS **112** selects and/or customizes a virtual-replica template may include, but are not limited to, model name, serial number, MAC addresses, device expansion module type, interface module type, and stacking composition. According to some aspects, the software characteristics based on which DTIS **112** customizes a virtual-replica template may include, but are not limited to, NOS type, NOS version, feature licenses, VLAN parameters, STP parameters, OSPF parameters, fabric parameters, VRRP parameters, and the like.

(32) According to some aspects, once the device-DTs **110***a*-**110***e* are instantiated, DTIS **112** may use the link connectivity information obtained from NMS **102** to generate virtual connections between the device-DTs to create network-DT **116**. The connectivity between device-DTs **110***a*-**110***e* may mirror the connectivity between the physical network devices **106***a*-**106***e* of the physical network **104**. According to some aspects, the instantiated virtual connection between two device-DTs (e.g., between device DTs **110***a* and **110***c*) may be selected to have a similar bandwidth and propagation characteristics as the physical connection between the counterpart physical network devices (e.g., between physical network devices **106***a* and **106***c*).

(33) At **222**, device-DT instances **110***a*-**110***e* may request additional device characteristics information from DTIS **112**. According to some aspects, at **222**, a base version of an instantiation of device-DT (e.g., device-DT **110***e*) may request additional hardware and/or software characteristics and/or connectivity information corresponding to its counterpart physical network device (i.e., physical network device **106***e*). According to some aspects, a device-DT instance may send a <get> RPC message to obtain device characteristics information from DTIS **112**.

(34) At **224**, DTIS **112** sends additional hardware and/or software characteristics corresponding to physical network device **204**. Device-DT **210** may be further customized based on the received

hardware and/or software characteristics. According to some aspects, NMS **102** and/or DTIS **112** may create a DT management channel between NMS **102** and each of the device-DTs of network-DT **116**, via DTIS **112**. According to some aspects, the management channel can be established using SSH or SSL/TSL protocols.

(35) At **226**, a device-DT of network-DT **116** (e.g., device-DT **110***e*) may request NMS **102** to send information corresponding to the current running configuration of its counterpart physical network device (e.g., physical network device **106***e*). A device-DT may communicate with NMS **102** using the management channel. At **228**, the requesting device-DT may receive an updated configuration bundle from NMS **102**, and the device-DT is further customized and/or updated based on the received running configuration parameters.

(36) FIG. **3** illustrates an example exchange of communications that may occur for synchronizing configuration changes between a network digital twin and a corresponding physical network, according to some aspects of this disclosure. In the illustrated example, the network system **300** includes a client device **114**, a physical network with physical network devices **106***a*-**106***e*, NMS **102**, DTIS **112**, and network-DT **116** that includes device-DTs **110***a*-**110***e*.

(37) At **312**, client device **114** sends a request to NMS **102** to initiate configuration changes to one or more device-DTs of network-DT **116**. Alternatively, or additionally, client device **114** may send a request to initiate configuration changes directly to one or more device-DTs of network-DT **116**. According to some aspects, client device **114** may want to validate an updated configuration on one or more device-DTs before implementing the updates at the counterpart physical network device. At **314**, NMS **102** forwards the configuration change request to the device-DTs using the management channel.

(38) At **316**, client device **114** provides NMS **102** with test configuration details for implementation and testing. At **318**, NMS **102** forwards the test configuration bundle to one or more device-DTs **110***a*-**110***e* of network-DT **116** using the management channel. According to some aspects, the configuration of one or more device-DTs **110***a*-**110***e* may be updated based on the received test configuration bundle. Furthermore, the control plane, management plane, and/or data plane operations of a physical network device (e.g., one of the physical network devices **106***a*-**106***e*) can be validated using its corresponding device-DT instance (e.g., one of the device-DTs **110***a*-**110***e*) with the updated configuration. The test configuration that is debugged and/or validated using network-DT **116** may subsequently be implemented at physical network **104**. According to some aspects, validating the test configuration includes running traffic through network-DT **116**.

(39) At **320**, the updated configuration of the device-DT instances is synchronized with a running configuration of the physical network devices in the physical network **104**. NMS **102** initiates the synchronization process and network-DT **116** sends the validated test configuration bundle to physical network devices **106***a*-**106***e*.

(40) At **322**, NMS **102** sends the validated test configuration to physical network **104**. The validated configuration is then applied to the corresponding physical network devices **106***a*-**106***e*. At **324**, NMS **102** may send a request to DTIS **112** to delete one or more device-DT instances (e.g., device-DTs **110***a*-**110***e*) corresponding to the physical network devices **106***a*-**106***e*. According to some aspects, in response to receiving the request to delete one or more device-DT instances, virtual links corresponding to the deleted device-DT instances are also destroyed within network-DT **116**. At **326**, the device-DT instance corresponding to the physical device **106***e* (e.g., device-DT **110***e*) is destroyed, and the processing resources are released.

(41) FIG. **4** is a conceptual diagram illustrating an exemplary method **400** to generate a device digital twin (device-DT) instance of a physical network device, according to some aspects of this disclosure. As a convenience and not a limitation, FIG. **4** may be described with regard to elements of FIGS. **1-3** and **6**. Method **400** may be performed by, for example, digital twin instantiation service (DTIS) **112**. Method **400** may also be performed by computer system **600** of FIG. **6**. But method **400** is not limited to the specific aspects depicted in those figures, and other systems may

be used to perform the method as will be understood by those skilled in the art. It is to be appreciated that not all operations may be needed, and the operations may not be performed in the same order as shown in FIG. **4**.

(42) At **402**, a request to generate a device digital twin (device-DT) instance corresponding to a physical network device located within physical network **104** is received from a network management system (NMS). For example, the DTIS (e.g., DTIS **112**) can receive the request to generate the device-DT instance corresponding to a physical network device (e.g., one of the physical network devices **106***a*-**106***e*). The request can include at least a first plurality and a second plurality of device characteristics corresponding to the physical network device. According to some aspects, the first plurality of device characteristics includes one or more of the hardware attributes of the physical network device including device model, device serial number, media access control (MAC) address, device expansion module type, or stacking composition.

(43) According to some aspects, DTIS **112** may store a plurality of virtual-replica templates corresponding to various types of physical network devices. Each virtual-replica template may define a superset of hardware and software attributes corresponding to a respective type of network device.

(44) At **404**, a virtual-replica template from a plurality of virtual-replica templates is selected based on the first plurality of device characteristics. For example, the DTIS (e.g., DTIS **112**) selects the virtual-replica template from the plurality of virtual-replica templates. According to some aspects, the first plurality of device characteristics may include one or more of the hardware attributes of the physical network device including device model, device serial number, media access control (MAC) address, device expansion module type, interface module type, or stacking composition.

(45) At **406**, once a suitable virtual-replica template is selected based on the first plurality of device characteristics (e.g., the device model, the serial number, or the like), the virtual-replica template is customized based on the second plurality of device characteristics. For example, the DTIS (e.g., DTIS **112**) customizes the virtual-replica template based on the second plurality of device characteristics. According to some aspects, the second plurality of device characteristics may include a running configuration of the physical network device, where the running configuration includes one or more software attributes including network operating system type, network operating system version, or feature licenses. According to some aspects, the second plurality of device characteristics includes limit memory and request memory values, where the limit memory and the request memory values may be provisioned based on the type and the model of the physical network device. Furthermore, the second plurality of device characteristics may include network connectivity information learned by the physical network device (e.g., one of the physical network devices **106***a*-**106***e*) through network discovery.

(46) At **408**, the device-DT instance of the physical network device is generated based on the customized virtual-replica template. According to some aspects, DTIS **112** may be configured to instantiate the customized virtual-replica template to generate a virtual replica of the corresponding physical network device.

(47) At **410**, a management channel is established between the device-DT instance (e.g., device-DT **110***a*-**110***e*) and the NMS. According to some aspects, the management channel is established, via the DTIS (e.g., DTIS **112**), between each device-DT (e.g., device-DTs **110***a*-**110***e*) and the NMS (e.g., NMS **102**). According to some aspects, a request to modify one or more device-DT configurations may be received, by the DTIS, from the NMS receive over the management channel. According to some aspects, the configuration of a device-DT instance may be updated based on the request to modify configuration, and the updated configuration of the device-DT instance and a running configuration of the corresponding physical network device are synchronized using the management channel. According to some aspects, a change in the configuration of a physical network device (e.g., one of the physical network devices **106***a*-**106***e*) may initiate a synchronization procedure between the physical network device and its

corresponding device-DT in the network-DT **116**. Similarly, a change in the configuration of a device-DT (e.g., one of the device-DTs **110***a*-**110***e*) may initiate a synchronization procedure between the device-DT and its corresponding physical network device in physical network **104**. According to some aspects, the control plane, management plane, and data plane operations of the physical network device may be validated using the device-DT with the updated configuration.
(48) FIG. **5** is a conceptual diagram illustrating an exemplary method **500** to generate a network digital twin (network-DT) instance of a physical network, according to some aspects of this disclosure. As a convenience and not a limitation, FIG. **5** may be described with regard to elements of FIGS. **1-3** and **6**. Method **500** may be performed by, for example, DTIS **112**. Method **500** may also be performed by computer system **600** of FIG. **6**. But method **500** is not limited to the specific aspects depicted in those figures, and other systems may be used to perform the method as will be understood by those skilled in the art. It is to be appreciated that not all operations may be needed, and the operations may not be performed in the same order as shown in FIG. **5**.
(49) At **502**, a request to generate a network digital twin (network-DT) instance corresponding to a physical network (e.g., physical network **104**) including a plurality of physical network devices (e.g., physical network devices **106***a*-**106***e*) is received from a network management system (NMS). For example, the DTIS (e.g., DTIS **102**) can receive the request to generate the network-DT instance. According to some aspects, the request includes link connectivity information corresponding to the physical network and a respective plurality of device characteristics corresponding to each of the plurality of physical network devices. Additionally, or alternatively, the DTIS (alone or in combination with the NMS) can determine the link connectivity information corresponding to the physical network and the respective plurality of device characteristics corresponding to each of the plurality of physical network devices from the physical network. For example, the DTIS (alone or in combination with the NMS) can "read" the link connectivity information and/or the respective plurality of device characteristics from the physical network
(50) At **504**, a plurality of device digital twin (device-DT) instances **110***a*-**110***e*, each corresponding to a respective physical network device of the plurality of physical network devices, are generated. For example, the DTIS can generate the device-DT instances. According to some aspects, each device-DT instance of the plurality of device-DT instances is generated based on the respective plurality of device characteristics corresponding to the respective physical network device.
(51) According to some aspects, to generate a device-DT instance of the plurality of device-DT instances, the DTIS (e.g., DTIS **112**) may select a virtual-replica template from a plurality of virtual-replica templates based on a first plurality of device characteristics of the respective plurality of device characteristics. DTIS **112** may then customize the virtual-replica template based on the second plurality of device characteristics of the respective plurality of device characteristics and generate the device-DT instance of the physical network device based on the customized virtualized-replica template.
(52) At **506**, the network-DT instance is generated based on the plurality of device-DT instances and the link connectivity information. According to some aspects, the link connectivity information may be obtained by the NMS (e.g., NMS **102**) by receiving layer **2** and layer **3** connectivity information that the physical network devices (e.g., physical network devices **106***a*-**106***e*) may learn through network discovery. According to some aspects, the physical network devices (e.g., physical network devices **106***a*-**106***e*) may obtain link connectivity information using traceroute, simple network management protocol (SNMP) and/or link layer discovery protocol (LLDP).
(53) According to some aspects, a management channel between each of network-DT instances **110***a*-**110***e* of network-DT **116** and the NMS is established. According to some aspects, the management channel is established, via the DTIS (e.g., DTIS **112**), between the network-DTs (e.g., network-DTs **110***a*-**110***e*) and the NMS (e.g., NMS **102**). According to some aspects, each of the individual DT-devices within the created network-DT instance **116** have an established communication channel with NMS **102**, via hyper text transfer protocol (HTTP), SSH, or RPC.

According to some aspects, a request to modify the network-DT configuration may be received, by the DTIS, from the NMS over the management channel. According to some aspects, the configuration of the network-DT instance may be updated based on the request to modify the configuration, and the updated configuration of the network-DT instance and a running configuration of the physical network (e.g., physical network **104**) are synchronized using the management channel. According to some aspects, a change in the topology and/or configuration of the physical network **104** may automatically initiate a synchronization procedure between the network-DT **116** and NMS **102**. According to some aspects, a change in the topology of physical network **104** may include the addition and/or removal of physical network devices and the addition and/or deletion of network interconnection links. For example, if physical network device **106***e* were removed from physical network **104**, the corresponding digital twin device **110***e* would be removed from network-DT **116**. Similarly, if a new physical network device were added to the physical network **104**, a corresponding device-DT instance would be generated in the network-DT **116**. According to some aspects, the control plane, management plane, and data plane operations of the physical network **104** may be validated using the network-DT with the updated configuration.
(54) Various aspects may be implemented, for example, using one or more well-known computer systems, such as computer system **600** shown in FIG. **6**. One or more computer systems **600** may be used, for example, to implement any of the aspects discussed herein, as well as combinations and sub-combinations thereof.
(55) Computer system **600** may include one or more processors (also called central processing units, or CPUs), such as a processor **604**. Processor **604** may be connected to a communication infrastructure or bus **606**.
(56) Computer system **600** may also include user input/output device(s) **603**, such as monitors, keyboards, pointing devices, etc., which may communicate with communication infrastructure **606** through user input/output interface(s) **602**.
(57) One or more of processors **604** may be a graphics processing unit (GPU). In an aspect, a GPU may be a processor that is a specialized electronic circuit designed to process mathematically intensive applications. The GPU may have a parallel structure that is efficient for parallel processing of large blocks of data, such as mathematically intensive data common to computer graphics applications, images, videos, etc.
(58) Computer system **600** may also include a main or primary memory **608**, such as random access memory (RAM). Main memory **608** may include one or more levels of cache. Main memory **608** may have stored therein control logic (i.e., computer software) and/or data.
(59) Computer system **600** may also include one or more secondary storage devices or memory **610**. Secondary memory **610** may include, for example, a hard disk drive **612** and/or a removable storage device or drive **614**. Removable storage drive **614** may be a floppy disk drive, a magnetic tape drive, a compact disk drive, an optical storage device, tape backup device, and/or any other storage device/drive.
(60) Removable storage drive **614** may interact with a removable storage unit **618**. Removable storage unit **618** may include a computer usable or readable storage device having stored thereon computer software (control logic) and/or data. Removable storage unit **618** may be a floppy disk, magnetic tape, compact disk, DVD, optical storage disk, and/any other computer data storage device. Removable storage drive **614** may read from and/or write to removable storage unit **618**.
(61) Secondary memory **610** may include other means, devices, components, instrumentalities or other approaches for allowing computer programs and/or other instructions and/or data to be accessed by computer system **600**. Such means, devices, components, instrumentalities or other approaches may include, for example, a removable storage unit **622** and an interface **620**. Examples of the removable storage unit **622** and the interface **620** may include a program cartridge and cartridge interface (such as that found in video game devices), a removable memory chip (such as an EPROM or PROM) and associated socket, a memory stick and USB port, a memory card and

associated memory card slot, and/or any other removable storage unit and associated interface.

(62) Computer system **600** may further include a communication or network interface **624**. Communication interface **624** may enable computer system **600** to communicate and interact with any combination of external devices, external networks, external entities, etc. (individually and collectively referenced by reference number **628**). For example, communication interface **624** may allow computer system **600** to communicate with external or remote devices **628** over communications path **626**, which may be wired and/or wireless (or a combination thereof), and which may include any combination of LANs, WANs, the Internet, etc. Control logic and/or data may be transmitted to and from computer system **600** via communication path **626**.

(63) Computer system **600** may also be any of a personal digital assistant (PDA), desktop workstation, laptop or notebook computer, netbook, tablet, smart phone, smart watch or other wearable, appliance, part of the Internet-of-Things, and/or embedded system, to name a few non-limiting examples, or any combination thereof.

(64) Computer system **600** may be a client or server, accessing or hosting any applications and/or data through any delivery paradigm, including but not limited to remote or distributed cloud computing solutions; local or on-premises software ("on-premise" cloud-based solutions); "as a service" models (e.g., content as a service (CaaS), digital content as a service (DCaaS), software as a service (SaaS), managed software as a service (MSaaS), platform as a service (PaaS), desktop as a service (DaaS), framework as a service (FaaS), backend as a service (BaaS), mobile backend as a service (MBaaS), infrastructure as a service (IaaS), etc.); and/or a hybrid model including any combination of the foregoing examples or other services or delivery paradigms.

(65) Any applicable data structures, file formats, and schemas in computer system **600** may be derived from standards including but not limited to JavaScript Object Notation (JSON), Extensible Markup Language (XML), Yet Another Markup Language (YAML), Extensible Hypertext Markup Language (XHTML), Wireless Markup Language (WML), MessagePack, XML User Interface Language (XUL), or any other functionally similar representations alone or in combination. Alternatively, proprietary data structures, formats or schemas may be used, either exclusively or in combination with known or open standards.

(66) In some aspects, a tangible, non-transitory apparatus or article of manufacture comprising a tangible, non-transitory computer useable or readable medium having control logic (software) stored thereon may also be referred to herein as a computer program product or program storage device. This includes, but is not limited to, computer system **600**, main memory **608**, secondary memory **610**, and removable storage units **618** and **622**, as well as tangible articles of manufacture embodying any combination of the foregoing. Such control logic, when executed by one or more data processing devices (such as computer system **600**), may cause such data processing devices to operate as described herein.

(67) Based on the teachings contained in this disclosure, it will be apparent to persons skilled in the relevant art(s) how to make and use aspects of this disclosure using data processing devices, computer systems and/or computer architectures other than that shown in FIG. **6**. In particular, aspects can operate with software, hardware, and/or operating system implementations other than those described herein.

(68) It is to be appreciated that the Detailed Description section, and not any other section, is intended to be used to interpret the claims. Other sections can set forth one or more but not all exemplary aspects as contemplated by the inventor(s), and thus, are not intended to limit this disclosure or the appended claims in any way.

(69) While this disclosure describes exemplary aspects for exemplary fields and applications, it should be understood that the disclosure is not limited thereto. Other aspects and modifications thereto are possible, and are within the scope and spirit of this disclosure. For example, and without limiting the generality of this paragraph, aspects are not limited to the software, hardware, firmware, and/or entities illustrated in the figures and/or described herein. Further, aspects (whether

or not explicitly described herein) have significant utility to fields and applications beyond the examples described herein.
(70) Aspects have been described herein with the aid of functional building blocks illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the description. Alternate boundaries can be defined as long as the specified functions and relationships (or equivalents thereof) are appropriately performed. Also, alternative aspects can perform functional blocks, steps, operations, methods, etc. using orderings different than those described herein.
(71) References herein to “one aspect,” “an aspect,” “an example aspect,” or similar phrases, indicate that the aspect described can include a particular feature, structure, or characteristic, but every aspect can not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same aspect. Further, when a particular feature, structure, or characteristic is described in connection with an aspect, it would be within the knowledge of persons skilled in the relevant art(s) to incorporate such feature, structure, or characteristic into other aspects whether or not explicitly mentioned or described herein. Additionally, some aspects can be described using the expression “coupled” and “connected” along with their derivatives. These terms are not necessarily intended as synonyms for each other. For example, some aspects can be described using the terms “connected” and/or “coupled” to indicate that two or more elements are in direct physical or electrical contact with each other. The term “coupled,” however, can also mean that two or more elements are not in direct contact with each other, but yet still co-operate or interact with each other.
(72) The breadth and scope of this disclosure should not be limited by any of the above-described exemplary aspects, but should be defined only in accordance with the following claims and their equivalents.

Claims

1. A system, comprising: a memory; and one or more processors coupled to the memory and configured to: receive, from a network management system (NMS), a request to generate a network digital twin (network-DT) instance corresponding to a physical network comprising a plurality of physical network devices, wherein the request includes link connectivity information corresponding to the physical network and a respective plurality of device characteristics corresponding to each of the plurality of physical network devices; generate a plurality of device digital twin (device-DT) instances each corresponding to a respective physical network device of the plurality of physical network devices, wherein each device-DT instance of the plurality of device-DT instances is generated based on the respective plurality of device characteristics corresponding to the respective physical network device, wherein to generate a device-DT instance of the plurality of device-DT instances, the one or more processors are further configured to: select a virtual-replica template from a plurality of virtual-replica templates based on a first plurality of device characteristics of the respective plurality of device characteristics; customize the virtual-replica template based on a second plurality of device characteristic of the respective plurality of device characteristics; and generate the device-DT instance of the physical network device based on the customized virtual-replica template; and generate the network-DT instance based on the plurality of device-DT instances and the link connectivity information.
2. The system of claim 1, wherein the one or more processors are further configured to: receive, over a management channel, a request to modify configuration of the network-DT from the NMS, wherein the request to modify the configuration includes a set of modified configuration parameters; update the configuration of the network-DT instance based on the modified configuration parameters; and synchronize, using the management channel, the updated configuration of the network-DT instance and a running configuration of the plurality of physical

network devices corresponding to the physical network.

3. The system of claim 2, wherein the one or more processors are further configured to: validate control plane, management plane, and data plane operations using the network-DT with the updated configuration.

4. The system of claim 1, wherein the first plurality of device characteristics comprises one or more of the following hardware attributes of the physical network device: device model, device serial number, media access control (MAC) address, and stacking composition.

5. The system of claim 1, wherein the second plurality of device characteristics comprises a running configuration of the physical network device and device configuration parameters of the physical network device, wherein the running configuration includes one or more of the following software attributes: network operating system type, network operating system version, and feature licenses.

6. The system of claim 1, wherein a configuration of the network-DT includes the link connectivity information corresponding to the physical network.

7. The system of claim 1, wherein the link connectivity information comprises identities of directly connected neighboring devices for each physical network device of the plurality of physical network devices and ports that are used for connecting to the neighboring devices.

8. A method for generating a network digital twin (network-DT) instance corresponding to a physical network comprising a plurality of physical network devices, the method comprising: receiving, from a network management system (NMS), a request to generate the network-DT instance corresponding to the physical network, wherein the request includes link connectivity information corresponding to the physical network and a respective plurality of device characteristics corresponding to each of the plurality of physical network devices; generating a plurality of device digital twin (device-DT) instances each corresponding to a respective physical network device of the plurality of physical network devices, wherein each device-DT instance of the plurality of device-DT instances is generated based on the respective plurality of device characteristics corresponding to the respective physical network device, wherein generating a device-DT instance of the plurality of device-DT instances further comprises: selecting a virtual-replica template from a plurality of virtual-replica templates based on a first plurality of device characteristics of the respective plurality of device characteristics; customizing the virtual-replica template based on a second plurality of device characteristic of the respective plurality of device characteristics; and generating the device-DT instance of the physical network device based on the customized virtual-replica template; and generating the network-DT instance based on the plurality of device-DT instances and the link connectivity information.

9. The method of claim 8, further comprising: receiving, over a management channel, a request to modify a configuration of the network-DT from the NMS, wherein the request to modify the configuration includes a set of modified configuration parameters; update the configuration of the network-DT instance based on the modified configuration parameters; and synchronize, using the management channel, the updated configuration of the network-DT instance and a running configuration of the plurality of physical network devices corresponding to the physical network.

10. The method of claim 9, further comprising: validating control plane, management plane, and data plane operations using the network-DT with the updated configuration.

11. The method of claim 8, wherein the first plurality of device characteristics comprises one or more of the following hardware attributes of the physical network device: device model, device serial number, media access control (MAC) address, and stacking composition.

12. The method of claim 8, wherein the second plurality of device characteristics comprises a running configuration of the physical network device and device configuration parameters of the physical network device, wherein the running configuration includes one or more of the following software attributes: network operating system type, network operating system version, and feature licenses.

13. The method of claim 8, wherein a configuration of the network-DT includes the link connectivity information corresponding to the physical network.
14. The method of claim 8, wherein the link connectivity information comprises identities of directly connected neighboring devices for each physical network device of the plurality of physical network devices and ports that are used for connecting to the neighboring devices.
15. A non-transitory computer-readable medium (CRM) having instructions stored thereon that, when executed by at least one computing device, cause the at least one computing device to perform operations comprising: receiving, from a network management system (NMS), a request to generate a network digital twin (network-DT) instance corresponding to a physical network comprising a plurality of physical network devices, wherein the request includes link connectivity information corresponding to the physical network and a respective plurality of device characteristics corresponding to each of the plurality of physical network devices; generating a plurality of device digital twin (device-DT) instances each corresponding to a respective physical network device of the plurality of physical network devices, wherein each device-DT instance of the plurality of device-DT instances is generated based on the respective plurality of device characteristics corresponding to the respective physical network device, wherein generating a device-DT instance of the plurality of device-DT instances comprises: selecting a virtual-replica template from a plurality of virtual-replica templates based on a first plurality of device characteristics of the respective plurality of device characteristics; customizing the virtual-replica template based on a second plurality of device characteristic of the respective plurality of device characteristics; and generating the device-DT instance of the physical network device based on the customized virtual-replica template; and generating the network-DT instance based on the plurality of device-DT instances and the link connectivity information.
16. The non-transitory CRM of claim 15, the operations further comprising: receiving, over a management channel, a request to modify a configuration of the network-DT from the NMS, wherein the request to modify the configuration includes a set of modified configuration parameters; updating the configuration of the network-DT instance based on the modified configuration parameters; and synchronizing, using the management channel, the updated configuration of the network-DT instance and a running configuration of the plurality of physical network devices corresponding to the physical network.
17. The non-transitory CRM of claim 16, the operations further comprising: validating control plane, management plane, and data plane operations using the network-DT with the updated configuration.
18. The non-transitory CRM of claim 15, wherein the first plurality of device characteristics comprises one or more of the following hardware attributes of the physical network device: device model, device serial number, media access control (MAC) address, and stacking composition.
19. The non-transitory CRM of claim 15, wherein the second plurality of device characteristics comprises a running configuration of the physical network device and device configuration parameters of the physical network device, wherein the running configuration includes one or more of the following software attributes: network operating system type, network operating system version, and feature licenses.
20. The non-transitory CRM of claim 15, wherein the link connectivity information comprises identities of directly connected neighboring devices for each physical network device of the plurality of physical network devices and ports that are used for connecting to the neighboring devices.